

Static testing technique with Example.

So static testing is a Software Testing technique which is used to check defects in Software application without executing the code. ~~Software~~ Static testing is done to avoid errors at an early stages of development as it is easier to Identify defects and solve the defects. It is also helps to finding errors that may not be found by Dynamic Testing.

The main two types of static testing technique are:

Manual examination:

Manual examination includes analysis of code done manually. It is also known as Reviews.

Automation analysis using tools:

Automation analysis are basically Static analysis which is done using tools.

Static testing

- Informal reviews.
- walkthrough
- Technical reviews
- Inspections
- Static Analysis

└ control flow
└ Data flow.

Static testing classified into 4 parts.

1. Informal reviews: Informal reviews are casual or unstructured assessment or feedback sessions typically occurring spontaneously between peers or stakeholders with minimal or no formal documentation.

2. walkthrough:

walkthrough is a step-by-step demonstration or explanation of a process or task, system which is used to guide a user review work.

3. Technical reviews:

To find defects and opportunities for improvement in a technical document or product like code.

4. Inspections:

Software Inspections are formal, static review process to identify defects in software artifacts.

Types of defects easily findout during static testing:

1. Deviation from standards
2. Design defect
3. Missing requirement

what is tested in static testing?

- unit Test case
- Business requirement Document
- system / functional requirement
- prototype
- prototype specification document
- Test case
- use cases (A use cases are the description how the user interacts with the system to achieve the specific goal)

unreachable code samples

public class unreachable {
 // unreachable code

Examples

void public static void main (String[] args)


```
System.out.println("This will be printed");  
return; // The execution is terminate here  
System.out.println("This will never be  
printed"); compiler error.
```

```
}  
  
}
```

```
public class unreachable example {  
    public static void main (String[] args);  
        for(;;) loop  
            System.out.println("Inside the loop");  
            break; // Exit the loop unconditionally  
        System.out.println("This will never be  
        printed"); compiler error.
```

Coding ³Standard example:

* coding structure and design:

* Modularity:

Break down the code into
smaller. function and resuability.

* Accessibility:

Ensuring the code is
written with accessibility in mind
especially for UI componants.

Static testing tools,

1. Soonts
2. lints
3. checkstyle
4. Sourcemeter.

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